

Description of Lab Work nº 1	
Course	Real-Time systems / <i>Sistemas de Tempo Real</i>
Year	2022/2023
Aim	Study concepts of Real-Time Systems and develop a project with a real-time kernel
Classes	5 classes x 3 hours + 15 extra hours (outside)
Delivery Date	29/10/2022
Concrete Objectives: - Interaction with physical systems (Interface boards, sensors, actuators, ...) - Interface board (Data acquisition) - Sensors - Actuators - Interface functions (in C++) - Apply the base real-time system's concepts, namely: - Task / Process: concurrency, - State, event, actions (triggered by events) - Synchronization and communication between tasks - Accessing shared resources / exclusive access resources - Using Real-Time kernels - freeRTOS - C/C++ Language - Solve the problem described in Annex 1.	

Annex 1 – Description of Work

This Lab-work aims at training the real-time concepts with a real-time kernel. The work comprises the development of a program that allows two or one player to play the Rooster Game (*Jogo do Galo*), using real-time approaches and a compact storage (3x3) where the players can put the crosses and circles.

The tools necessary to develop the work comprise a real-time kernel (freeRTOS), the Visual Studio 2019 C/C++ compiler and the hardware kit that represents the automated storage system, illustrated in fig. 1. To enhance students' performance other tools are available for developing this work, such as: a compact storage simulator, fig.2 and a set of low-level functions for software/hardware interaction, provided in the lab slides.

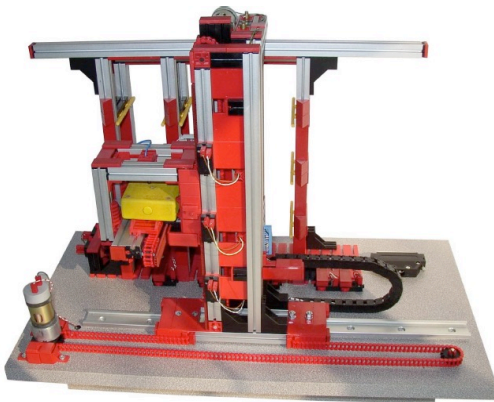


Fig. 1 – Automated Storage Kit

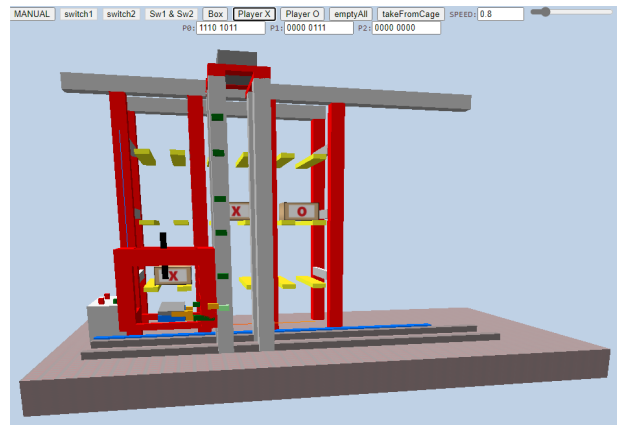


Fig. 2 – Automated Storage Simulator

The interaction between the real-time program and the automated storage is performed through an input / output digital data acquisition board, specifically, the NI USB 6509, as shown in fig. 3.



Fig. 3 – the NI USB 6509 data acquisition board

The **Automated Cyber-Physical Rooster Game** should consider the following aspects:

- Each part (Cross or Circle part) must be stored with the player who played, turn (*jogada*), and the date the move took place.
- There are only 9 storage cells available in the warehouse (3x3), so the system must know when the position is not valid. When this happens, the player loses the turn, so the other player must play.
- Each cell can hold a single part only.
- Parts (Circles or Crosses) are received for storage and delivered in position (1,1) with cage out ($y=3$).
- When moving along (x,z) , the cage should be in position ($y=2$).
- The game can be played by two human players, one human against computer or computer against computer.
- An alert should be triggered whenever a player wins the game (three parts in a row).
- An alert should be triggered whenever a player expires the time to play. The alert consists of the lamp (at the left) flashing.
- When a player takes more than X seconds to play, a random part already existent in the storage system, from that player is removed, and the player loses the turn.

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The functional **requirements** to develop the **Automated Cyber-Physical Rooster Game** are listed in the following table. They are split into basic or low-level requirements, for direct interaction with sensors and actuators, and high-level requirements, for storage management.

Table 1 – Needed requirements

Req.	Description
	Storage utilization requirements
FR1	Using the keyboard, develop a robust (semi-automatic) storage calibration. Using keys, the user defines the direction of the movements. During a movement, the actuator must stop when it reaches a position sensor.
FR2	Store a part in the storage. The system will store the part with the current date, player, and cell(x,y). The system should verify if the cell is empty.
FR3	Remove part from cell (x,z) specified through keyboard.
FR4	When starting a new game, the user must be able to select the difficulty of the game (time available to play). Easy = 10 seconds, Hard = 5 seconds.
FR5	The system must allow three types of games: 1. Normal game: The users can only add parts. 2. Hybrid game: The users can add or remove a part from the system (the player can remove one part from the opponent). 3. Random game. When the users get their turn, the system will say if the user can add or remove a part (randomly).
FR6	Activate “flashing lamp” (at the left) to alert the player expired the time available to play and lose turn.
FR7	When the player loses turn because of time expiring (FR6), the system should automatically remove one of the parts already played by this player (randomly). The “flashing lamp” (at the left) must be activated until the part is removed.
FR8	While being operated (moving), if both buttons are pressed, then perform emergency

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	STOP.
FR9	During an emergency STOP, if left button is pressed, then RESUME operation.
FR10	During an emergency STOP, if right button is pressed, then RESET the system (perform semi-automatic).
FR11	Both lights flash faster (0.5 seconds period) during an emergency STOP (and until resume).
FR12	The system must be able to have human and computerized players. So, in the end we can have games like: 1. Human – Human 2. Human – Computer 3. Computer - Computer
	Information retrieval requirements
FR13	Show list of stored parts (type, player, turn and date when played).
FR14	Given a part type (crosses and circles), shows all corresponding (x,z) coordinates.
FR15	Show the record (historical) of all parts removed from the storage.
	Non-Functional requirements:
NF1	While operating, the system can provide information (e.g., occupation level / free cells / part list).
NF2	The system works appropriately (one part per cell, not taking parts from empty cells)
NF3	The system always moves within limits x, y, z.
NF4	The system moves along (x,z) only when y is stopped at center position ($y=2$) .
NF5	The system delivers and receives products with y out ($y=3$) in coordinate $(x,y)=(1,1)$

Classes plan for the Lab Work 1

Class 1: [22 – 26 September 2022]

SW Installation & Interaction with “physical” system.

Class 2: [29 September – 3 October 2022]

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Interaction between a program and physical systems, through sensors, actuators, and a data acquisition board; Development of the functions for the interaction with the warehouse.

Class 3: [6 – 10 October 2022]

Development of real-time behaviour; tasks and semaphores; development of the low-level requirements.

Class 4: [13 – 17 October 2022]

Communication mechanisms (mailboxes); development of the high-level requirements to control the warehouse.

Class 5: [20 – 24 October 2022]

Conclusion of LabWork1.

Professors:

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Lab Work 1 Delivery:

Students must submit via Moodle (**and only via Moodle**):

- ➔ Upload a zip/rar archive with your Visual Studio Project.
- ➔ Upload a demo video (aprox. 5/8 minutes) presenting all implemented requirements/functionality. The video **must be** accompanied with a narration of what is being presented.

Evaluation criteria

Items			Quotation/ Penalty
Functionality	FR1	Using the keyboard, develop a robust (semi-automatic) storage calibration. Using keys, the user defines the direction of the movements. During a movement, the actuator must stop when it reaches a position sensor.	0.5
	FR2	Store a part in the storage. The system will store the part with the current date, player, and cell(x,y). The system should verify if the cell is empty.	0.5
	FR3	Remove part from cell (x,z) specified through keyboard.	1
	FR4	When starting a new game, the user must be able to select the difficulty of the game (time available to play). Easy = 10 seconds, Hard = 5 seconds.	0.5
	FR5	The system must allow three types of games: 4. Normal game: The users can only add parts. 5. Hybrid game: The users can add or remove a part from the system (the player can remove one part from the opponent). Random game. When the users get their turn, the system will say if the user can add or remove a part (randomly).	1
	FR6	Activate “flashing lamp” (at the left) to alert the player expired the time available to play and lose turn.	1
	FR7	When the player loses turn because of time expiring (FR6), the system should automatically remove one of the parts already played by this player (randomly). The “flashing lamp” (at the left) must be activated until the part is removed.	1
	FR8	While being operated (moving), if both buttons are pressed, then perform emergency STOP.	1
	FR9	During an emergency STOP, if left button is pressed, then RESUME operation.	0.75
	FR10	During an emergency STOP, if right button is pressed, then RESET the system (perform semi-automatic).	0.75
	FR11	Both lights flash faster (0.5 seconds period) during an emergency STOP (and until resume).	0.5
	FR12	The system must be able to have human and computerized players. So, in the end we can have games like: 4. Human – Human 5. Human – Computer Computer - Computer	2
	FR13	Show list of stored parts (type, player, turn and date when played).	0.5
	FR14	Given a part type (crosses and circles), shows all corresponding (x,z) coordinates.	0.5
	FR15	Show the record (historical) of all parts removed from the storage.	1
	NF1	While operating, the system can provide information (e.g., occupation level / free cells / part list).	2
	NF2	The system works appropriately (one part per cell, not taking parts from empty cells)	2
	NF3	The system always moves within limits x, y, z.	0.5
	NF4	The system moves along (x,z) only when y is stopped at center position (y==2) .	0.5
	NF5	The system delivers and receives products with y out (y=3) in coordinate (x,y)=(1,1)	0.5
Video			2
TOTAL			20
No delivery of the VS19 project or the video (* MANDATORY)			-20